COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 | Data Structures and Algorithms	**SECTION:**	CPE21S3
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Hands-on Activity 9.1 Tree ADT

1. PROCEDURE OUTPUT

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Step 1: Implement a binary search tree that will take the following values: 2, 3, 9, 18, 0, 1, 4, 5.

```
22  template<typename T>
23  BinarySearchTree<T>::BinarySearchTree(T value){
24      key = value;
25      left = nullptr;
26      right = nullptr;
27  }
28
29  template<typename T> // recursive calling
30  void BinarySearchTree<T>::insert(T value){
31      //go to left if : value < key
32      if(value < key){
33          if(left == nullptr){
34              left = new BinarySearchTree<T>(value);
35          }else{
36              //go back to start so the value on this wi
37              left->insert(value);
38          }
39      }
40
41      //go to right if : value > key
42      else if( value > key){
43          if(right == nullptr){
44              right = new BinarySearchTree<T>(value);
45          }else{
46              //go back to start so the value on this wi
47              right->insert(value);
48          }
49      }
50  }
```

```cpp
#include <iostream>
#include "BST.h"

int main(){

    BinarySearchTree<int>* tree = new BinarySearchTree(2);

    tree->insert(3);
    tree->insert(9);
    tree->insert(18);
    tree->insert(0);
    tree->insert(1);
    tree->insert(4);
    tree->insert(5);
```

Step 2: Create a diagram to show the tree after all values have been inserted. Then, with the use of visual aids (like arrows and numbers) indicate the traversal order for in-order, pre-order and post-order traversal on the diagram.

Step 3: Compare the different traversal methods. In-order traversal was performed with what function?

In-order traversal

`std::cout << "In Order Output: ";`
`tree->inOrder();`
`std::cout << std::endl;`

`template<typename T>`
`void BinarySearchTree<T>::inOrder(){`
`  if(left != nullptr){`
`  //recursive calling`
`  left->inOrder();`
`  }`
`  std::cout << key << " ";`
`  if(right != nullptr){`
`  //recursive calling`
`  right->inOrder();`
`  }`
`}`

`In Order Output: 0 1 2 3 4 5 9 18`

Pre-order Traversal

```
std::cout << "Pre Order Output: ";
tree->preOrder();
std::cout << std::endl;
```

```
template<typename T>
void BinarySearchTree<T>::preOrder(){
    std::cout << key << " ";
    if(left != nullptr){
        //recursive calling
        left->preOrder();
    }
    if(right != nullptr){
        //recursive calling
        right->preOrder();
    }
}
```

```
Pre Order Output: 2 0 1 3 9 4 5 18
```

Post-order Traversal

```
std::cout << "Post Order Output: ";
tree->postOrder();
std::cout << std::endl;
```

```cpp
template<typename T>
void BinarySearchTree<T>::postOrder(){
    if(left != nullptr){
        //recursive calling
        left->postOrder();
    }

    if(right != nullptr){
        //recursive calling
        right->postOrder();
    }

    std::cout << key << " ";
}
```

```
Post Order Output: 1 0 5 4 18 9 3 2
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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